Annotated bibliography for the almost-all non-halting theorem

BB(6) research program

2026-09-08

This bibliography accompanies the literature review and the manuscript. Each entry states the reference, the precise result, and its role in the proof. Bibliographic details were verified against primary sources (publisher pages, NUMDAM, AMS, IMPAN, Mathnet, and in two cases the article PDFs directly); residual uncertainties are flagged in *italics*. Nothing here is cited for an effect its source does not actually provide — in particular we distinguish what supplies the ACIM from what supplies the *rate*, and what is finite-measure from what is infinite-measure.

1 Foundations and the Collatz template

Tao (2022). *Almost all orbits of the Collatz map attain almost bounded values*, Forum of Mathematics, Pi **10**, e12 (arXiv:1909.03562). The theorem: for any $f \to \infty$, $\mathrm{Col}_{\min}(N) < f(N)$ for almost all N in *logarithmic* density. The method passes to the Syracuse (odd-to-odd) map, models the 2-adic valuations by a "Syracuse random variable," and proves a local limit theorem controlled by a bound on the characteristic function of a skew random walk on $\mathbb{Z}/3^n\mathbb{Z}$. **Role:** the template for the entire program — reduce the arithmetic map, model the increments, prove a local-limit/equidistribution estimate, conclude an almost-all statement. We transcribe its architecture (statistical model + quantitative rate; first-passage decomposition) while replacing its 3-adic harmonic analysis by a suspension/Dolgopyat argument, since our cost is multiplicative rather than a Birkhoff sum on a compact group.

Korec (1994); Krasikov–Lagarias (2003). *A density estimate for the* $3x+1$ *problem*, Math. Slovaca **44**, 85–89; and *Bounds for the* $3x{+}1$ *problem using difference inequalities*, Acta Arith. **109**, 237–258 (arXiv:math/0205002). The natural-density predecessors: Korec bounds the eventual minimum below N^θ, $\theta > \log 3/\log 4$, for almost all N; Krasikov–Lagarias show $\geq x^{0.84}$ of integers below x provably reach 1. **Role:** they calibrate "almost all" — these are power-law partial results, and Tao's advance (which we follow) was to reach density one via a genuine LLT rather than a sieve. *Applegate–Lagarias (Math. Comp. 64, 1995) and Lagarias's annotated* $3x{+}1$ *bibliography are companion pointers; exact pages/arXiv ids there are medium-confidence.*

Lagarias–Weiss (1992). *The* $3x+1$ *problem: two stochastic models*, Ann. Appl. Probab. **2**, 229–261. The rigorous stochastic heuristics that the density results approximate. **Role:** the ancestor of the "biased random walk" intuition that our theorem makes rigorous for the renewal map. *Pages high-confidence; confirm before print.*

Aaronson (2020). *The Busy Beaver Frontier*, ACM SIGACT News **51**(3), 2020 (also ECCC TR20-115). A survey of BB(n): growth, independence from ZFC of BB(n) for explicit n, and

small machines encoding open conjectures. **Role:** frames the meta-mathematical stakes and the bridge from BB(6) machines to Collatz-type dynamics. *Pages 32–54 medium-confidence.*

The bbchallenge Collaboration (2024–2025). The Coq-verified proof that BB(5) = 47,176,870 (Coq-BB5, led by the contributor "mxdys," announced July 2024; companion paper arXiv:2509.12337), and *Antihydra*, the six-state machine whose halting is equivalent to a Collatz-type first-passage event for $n \mapsto \lfloor 3n/2 \rfloor$ from $n = 8$. **Role:** the concrete BB(6) context — Antihydra is the canonical exhibit of a machine whose single-orbit behaviour is Collatz-hard while its statistics are the natural object of an almost-all theorem. *Authorship is collective/ pseudonymous; the halting-probability figure is a heuristic, not a theorem; the current* BB(6) *champion value should be quoted from the bbchallenge wiki if used. As of 2025, Tao (2022) stands essentially alone as a rigorous almost-all-log-density result for a Collatz-type map; no credible extension to another arithmetic system was found — so a port to a* BB(6)*/Antihydra-derived map would be the first.*

2 Invariant measures, mixing, and the transfer operator

Lasota–Yorke (1973). *On the existence of invariant measures for piecewise monotonic transformations*, Trans. AMS **186**, 481–488. The Lasota–Yorke inequality $\mathrm{Var}(\mathcal{L}f) \le \alpha\,\mathrm{Var}(f) + C\|f\|_1$, $\alpha < 1$, giving an ACIM with BV density. **Role:** the ancestor of every LY estimate we use.

Rychlik (1983). *Bounded variation and invariant measures*, Studia Math. **76**, 69–80. Extends Lasota–Yorke to *countably many* branches under summable variation: unique ACIM, BV density, quasi-compact transfer operator. **Role:** this is what makes our countable-branch map's ACIM and untwisted spectral gap free — the accumulation point ρ^* is a point of strong expansion, so no infinite-measure machinery is needed. *Attribution caution: Rychlik supplies the countable-branch ACIM and BV framework; the exponential decay rate is classically co-attributed to Hofbauer–Keller (below), which we cite for the rate.*

Hofbauer–Keller (1982). *Ergodic properties of invariant measures for piecewise monotonic transformations*, Math. Z. **180**, 119–140. Exponential decay of correlations for piecewise-expanding interval maps. **Role:** the mixing-*rate* citation companion to Rychlik. *Pages high-confidence; not DOI-verified in this pass.*

Keller–Liverani (1999). *Stability of the spectrum for transfer operators*, Ann. Scuola Norm. Sup. Pisa (4) **28**, 141–152. Isolated spectral data are stable under a perturbation small in a weak norm (need not be small in the strong BV norm). **Role:** the theorem that transfers the finite-branch gap to the countable map via truncation, and that absorbs the $O(1/a)$ model-to-exact perturbation of the *bulk* branches. (Not the tool for the native-resolution scale perturbation — see the manuscript's item-1 discussion — which is handled instead by applying the gap per exact-scale map.)

Baladi (2018). *Dynamical Zeta Functions and Dynamical Determinants for Hyperbolic Maps: A Functional Approach*, Ergebnisse der Mathematik 3. Folge **68**, Springer. The standard modern account of transfer-operator spectral theory via anisotropic spaces. **Role:** background reference for the functional-analytic framework. *Note: a maps reference; it does not contain the Dolgopyat oscillatory-cancellation estimate, which is separate.*

Mayer (1990). *On the thermodynamic formalism for the Gauss map*, Comm. Math. Phys. **130**, 311–333. Transfer-operator spectral theory of the continued-fraction Gauss map. **Role:** the

Gauss-type exemplar our induced map is modelled on. *If a continued-fraction/modular equidistribution statement were wanted instead, Series (JLMS 1985) is the alternative; the plain Gauss–Kuzmin–Wirsing spectral picture (Wirsing, Acta Arith. 24, 1974) is the lighter option.*

3 The Dolgopyat method: uniform spectral gaps

Dolgopyat (1998). *On decay of correlations in Anosov flows*, Annals of Math. (2) **147**, 357–390 (verified from the article PDF). Introduces the "Dolgopyat method": an L^2 cancellation estimate on twisted transfer operators along the imaginary frequency, driven by uniform non-integrability (UNI) of the temporal-distance function, giving exponential mixing. **Role:** the origin of the uniform large-frequency gap we need.

Dolgopyat (1998), Prevalence of rapid mixing. Ergodic Theory Dynam. Systems **18**(5), 1097–1114. States UNI in reusable *symbolic* form as a checkable condition on the roof cocycle. **Role:** the precise citation for "UNI as a hypothesis one verifies," as opposed to the Annals paper's flow-specific use.

Naud (2005). *Expanding maps on Cantor sets and analytic continuation of zeta functions*, Ann. Sci. ENS (4) **38**, 116–153 (verified from the article PDF). **The best-matched reference.** For uniformly expanding Markov maps with real-analytic roof, the non-local integrability condition NLI (Def. 2.1) is that the temporal-distance function $\varphi_{\xi,\eta}$ have non-vanishing derivative somewhere — in the analytic case simply $\varphi_{\xi,\eta} \not\equiv 0$, i.e. non-proportional branch roofs. Theorem 2.3: under NLI, $\|\mathcal{L}_s^n\|_{C^1} \le C|\operatorname{Im} s|^{1+\epsilon}\rho^n$, $\rho < 1$, *uniformly in* $\operatorname{Im} s$; Theorem 1.7: a zero-free strip for the zeta function. **Role:** our UNI verification — branch forms $L_k(\xi) = (r_k - p_k) + (s_k - q_k)\xi$ pairwise non-proportional (slopes $-\rho_k$ distinct) — is exactly Naud's $\varphi_{\xi,\eta} \not\equiv 0$ on our arithmetic, so Theorem 2.3 is the gap we invoke.

Stoyanov (2011). *Spectra of Ruelle transfer operators for Axiom A flows*, Nonlinearity **24**, 1089–1120 (arXiv:0810.1126). The higher-dimensional Axiom-A-flow analogue of Naud's estimate under a UNI/Diophantine temporal-distance condition. **Role:** corroborating generality. *Several closely-titled Stoyanov papers exist; pin by arXiv id.*

Avila–Gouëzel–Yoccoz (2006). *Exponential mixing for the Teichmüller flow*, Publ. Math. IHÉS **104**, 143–211. Exponential mixing via a Dolgopyat estimate over a *countable-alphabet* (Rauzy–Veech) roof. **Role:** the methodological precedent that the Dolgopyat estimate survives countably many branches — moral support for our §(infinite) reduction.

Ruelle (1976); Pollicott (1985); Parry–Pollicott (1990). *Zeta-functions for expanding maps and Anosov flows*, Invent. Math. **34**, 231–242; *On the rate of mixing of Axiom A flows*, Invent. Math. **81**, 413–426; *Zeta functions and the periodic orbit structure of hyperbolic dynamics*, Astérisque **187–188**. The transfer-operator/zeta framework, Pollicott–Ruelle resonances, and the RPF/non-lattice theory. **Role:** the framework citations; Naud invokes Parry–Pollicott Thm 6.3 for the non-lattice continuation. *Pollicott 1985 pages secondary-verified.*

4 Local limit theorems

Nagaev (1957); Guivarc'h–Hardy (1988). *Some limit theorems for stationary Markov chains*, Theory Probab. Appl. **2**, 378–406; and *Théorèmes limites pour une classe de chaînes*

de Markov et applications aux difféomorphismes d'Anosov, Ann. IHP Probab. Statist. **24**(1), 73–98. The origin of the perturbed-operator (eigenvalue) approach to the CLT, and its adaptation to hyperbolic dynamics with LLT and renewal theorems. **Role:** the "origin" citations for the method that turns our spectral gap into a local limit theorem.

Hennion–Hervé (2001). *Limit Theorems for Markov Chains and Stochastic Properties of Dynamical Systems by Quasi-Compactness*, Springer LNM **1766**. The definitive treatment of the Nagaev–Guivarc'h method: CLT/LLT/Berry–Esseen from quasi-compactness plus analytic perturbation of the leading eigenvalue $\lambda(i\xi)$. **Role:** the machinery citation bridging our BV/Lipschitz gap to the LLT output.

Aaronson–Denker (2001). *Local limit theorems for partial sums of stationary sequences generated by Gibbs–Markov maps*, Stoch. Dyn. **1**(2), 193–237. LLT for Birkhoff sums of Lipschitz observables over Gibbs–Markov (countable-partition, big-image, bounded-distortion) maps, under aperiodicity (non-lattice) and finite variance. **Role:** the off-the-shelf LLT for our log-growth cocycle over the induced Gauss-type map. *Cite this long published title, not the short-title preprint.*

5 Countable-branch thermodynamic formalism (context, not load-bearing)

Sarig (1999). *Thermodynamic formalism for countable Markov shifts*, Ergodic Theory Dynam. Systems **19**(6), 1565–1593. Gurevich pressure, variational principle, and RPF theory for countable-state shifts under the BIP condition. **Role:** the symbolic backdrop for coding our countable-branch map; not load-bearing given Rychlik.

Aaronson–Denker–Urbański (1993). *Ergodic theory for Markov fibred systems and parabolic rational maps*, Trans. AMS **337**(2), 495–548. Thermodynamic formalism for countable Markov fibred systems. **Role:** structural context for the fibred/countable-branch setting.

Sarig (2002); Gouëzel (2004); Melbourne–Terhesiu (2012). *Subexponential decay of correlations*, Invent. Math. **150**, 629–653; *Sharp polynomial estimates for the decay of correlations*, Israel J. Math. **139**, 29–65; *Operator renewal theory and mixing rates for dynamical systems with infinite measure*, Invent. Math. **189**, 61–110. Operator-renewal theory for slow (polynomial/subexponential) mixing — Sarig/Gouëzel finite-measure, Melbourne–Terhesiu infinite-measure. **Role:** cited only to say they are *not* needed: our tail is geometric and our measure finite, so the truncation is elementary. Included so a reader is not left wondering whether the countable branches force slow-mixing machinery. *They do not.*

On confidence

Two load-bearing items (Dolgopyat 1998; Naud 2005) were verified from the article PDFs, including the theorem statements used. All other primary entries have venue/volume/year confirmed against publisher or archive pages; italicized flags mark page ranges or identifiers confirmed only from consistent secondary metadata, and the handful of companion references not separately DOI-checked. The two anomalous "post-Tao extension" preprints surfaced during the search were judged unverifiable and are deliberately *excluded*.